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MILITARY SPECIFICATION

MONITOR, PUMP NOISE, AN/WSQ-3 (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the requirements and tests for equipment intended to monitor the noise produced by pumps.

1.2 **Classification.** Pump noise monitors covered by this specification shall be of the following types, as specified (see 6.2):

a. *Type I.* Type I equipment shall consist of one control-monitor (see 3.5) and six type I transducers (see 3.4) complete with transducer mounting hardware (see 3.4.7) and mating cable connectors (see 3.4.6).

b. *Type II.* Type II equipment shall consist of one control-monitor (see 3.5), six type II transducers (see 3.4) complete with transducer mounting hardware (see 3.4.7) and mating cable connectors (see 3.4.6).

c. *Type III.* Type III equipment shall consist of one control-monitor (see 3.5), four type III transducers (see 3.4) complete with transducer mounting hardware (see 3.4.7) and mating cable connectors (see 3.4.6).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6665

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-S-901	Shock Tests, HI (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-2036	Enclosures for Electric and Electronic Equipment
MIL-C-5015	Connectors, Electrical, Circular Threaded, AN Type, General Specification for
MIL-M-10304	Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for
MIL-E-16400	Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts), Packaging of
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated
MIL-C-24643	Cables and Cords, Electrical, Low Smoke, for Shipboard Use, General Specification for
MIL-C-24643/32	Cable, Electrical, Type LS2SWAU (Including Variation LS2SWA)

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STANDARDS

MILITARY

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-242	Electronic Equipment Parts and Selected Standards
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-763	Locking Devices
MIL-STD-781	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
MIL-STD-1399 Section 300	Interface Standard for Shipboard Systems Electric Power, Alternating Current, (Metric)
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 **Non-Government publications.** The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 National Electric Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 **General.** The equipment shall be in accordance with MIL-E-16400 to the extent specified.

3.3 **Environmental conditions.** All of the requirements specified in 3.4 through 3.11.4 shall be met under any combination of the environmental conditions specified in 3.3.1 through 3.3.8 and the steady state voltage and frequency tolerances specified in 3.7.1.

3.3.1 **Temperature conditions.** The equipment shall conform to the operating and non-operating temperature ranges as specified in table I.

TABLE I. *Temperature conditions.*

Equipment	Operating range	Nonoperating range
Control-monitor	5 to 50 °C	−62 to +71 °C
Transducers	5 to 100 °C	−62 to +100 °C

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3.3.2 **Humidity.** The equipment shall operate at relative humidities up to 95 percent for both continuous and intermittent periods, including conditions wherein condensation takes place in and on the equipment.

3.3.3 **Incident nuclear radiation.** For their design life, the transducers shall operate under the following continuous incident radiation:

Thermal neutrons	Up to 5×10^7 neutrons/cm^2 sec
Gamma irradiation	Up to 500 radius/hr

3.3.4 **Pressure conditions.** The equipment shall operate without damage or failure when subjected to an absolute pressure of 14.7 pounds per square inch (lb/in^2) (1.01×10^4 pascal (Pa)) with variations from a maximum of 18.7 lb/in^2 (1.29×10^5 Pa) to a minimum of 11.7 lb/in^2 (8.07×10^4 Pa). The equipment shall operate without damage or failure when subjected to momentary pressure excursions up to 30 lb/in^2 (2.07×10^5 Pa) and down to 10 lb/in^2 (6.89×10^4 Pa). The maximum allowable rated operating temperature (not the derated value) of each component part shall not be exceeded when subjected to a continuous absolute pressure of 11.7 lb/in^2 (1.01×10^4 Pa).

3.3.5 **Inclination.** The equipment shall not be damaged or caused to malfunction when permanently inclined at 60 degrees from the vertical in any direction (front, back, right, or left). No favorable orientation may be assumed.

3.3.6 **Roll and pitch.** The equipment shall not be damaged or caused to malfunction when subjected to values of roll and pitch up to 60 degrees from the vertical. No favorable orientation may be assumed.

3.3.7 **Vibration.** The equipment shall conform to the requirements of MIL-STD-167-1 for type I environmental vibration.

3.3.8 **Shock.** The equipment shall conform to the requirements of MIL-S-901 for grade A, type A, class I, lightweight equipment.

3.4 **Transducers.** One transducer shall be used to monitor the noise of one pump. Each transducer shall be a self-generating type with separate accelerometers producing electrical outputs whose amplitudes are proportional to the acceleration of pump vibration in the direction monitored by that accelerometer. Type I and type III transducers shall contain three separate accelerometers positioned to monitor vibration directed along three axes, oriented as follows:

a. Perpendicular to the mounting face of the transducer

b. Parallel to the long axis of the transducer

c. Parallel to the mounting face of the transducer and perpendicular to the long axis of the transducer.

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Type II transducers shall contain two accelerometers positioned to monitor vibrations directed along two axis, oriented as follows:

a. Perpendicular to the mounting face of the transducer
b. Parallel to the long axis of the transducer.

3.4.1 Sensitivity. Each crystal accelerometer shall be sensitive to vibrations having accelerations ranging from minus 30 to plus 20 decibels (dB) relative to 1 g (where g is acceleration due to gravity and 1 g is equal to 9.81 meters (m) (32 feet) per second per second), over the frequency range of 30 to 15,000 hertz (Hz). The output of each transducer shall be sufficient over the specified frequency range when it is subjected to acceleration in any direction in which it monitors vibration (see 3.4) to enable the control-monitor (see 3.5) to conform to the requirements of 3.5.2.2 and 3.5.2.3 when the transducer and control-monitor are connected as specified.

a. Type I transducers shall be connected to the control-monitor by one active shielded pair of type LS2SWA cable 45.72 m (150 feet) in length. Type LS2SWA cable shall be as specified in MIL-C-24643 and MIL-C-24643/32.

b. Type II transducer shall have the two accelerometers connected in parallel at an external junction box. This parallel combination shall be connected to the control-monitor by one active shielded pair of type LS2SWA cable 45.72 m (150 feet) in length.

c. Type III transducers shall have two of the three accelerometers connected in parallel at an external junction box. This parallel combination shall be connected to the control-monitor by one active shielded pair of type LS2SWA cable 45.72 m (150 feet) in length.

3.4.2 Capacitance. Each crystal accelerometer in all types of transducers shall have a capacitance of 1,000 picofarads (pF) plus or minus 20 percent.

3.4.3 Lock-wiring. Bolts used to hold a removable wall of the transducer case to the main transducer case shall be lock-wired in accordance with MIL-STD-763.

3.4.4 Transducer wiring. Internal wiring of all type transducers shall permit, by external selection of connector terminals, the choice of ground or no-ground for the twisted-pair shields. Selection of the connector and pin connections shall be subject to approval by the contracting activity.

3.4.4.1 Type I wiring. The three crystal accelerometers shall be wired in parallel. The connector used shall be a five-pin type in accordance with MIL-C-5015. One pin shall provide for internal ground and two pins shall have no internal connection.

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3.4.4.2 **Type II wiring.** Two crystal accelerometers shall be wired to provide completely separate outputs using four pins of the transducer connector. The connector used shall be a five-pin type in accordance with MIL-C-5015. One separate pin shall provide for internal ground in the transducer.

3.4.4.3 **Type III wiring.** The three crystal accelerometers shall be wired to provide completely separate outputs using six pins of the transducer connector. The connector used shall be a nine-pin type in accordance with MIL-C-5015. One separate pin shall provide for internal ground of the transducer and two pins shall have no internal connection.

3.4.5 **Materials.** The transducer enclosure, exclusive of the integral portion of the connector, shall be made of austenitic corrosion-resistant steel.

3.4.5.1 **Recovered materials.** Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4.6 **Cable connector.** A cable connector in accordance with MIL-C-5015 shall be furnished with each transducer. This connector shall mate with the connector integral with the transducer (see 3.5.2.2) and shall be capable of installation on type LS2SWA cable.

3.4.7 **Transducer mounting hardware.** Four austenitic corrosion-resistant steel 6.35-millimeter (mm) (0.25-inch) UNC2A bolts shall be furnished with each transducer. An austenitic corrosion-resistant steel locking device in accordance with MIL-E-16400 shall be furnished with each bolt. Each bolt head shall be drilled to accept lock-wiring in accordance with MIL-STD-763. The lengths of bolts shall be such as to penetrate into the mounting surface of the pump not more than 12.70 mm (0.50-inch)

3.5 **Control-monitor.** The control-monitor shall consist of a pump selector switch, an amplifier, a headset circuit, a calibration circuit, and a display circuit. The display may consist of a digital numeric display or a meter.

3.5.1 **Pump selector switch.** A pump selector switch shall be provided on the front panel to permit selection of any one of six transducer outputs as the input to the amplifier. These six positions of this switch shall be labeled from 1 to 6, consecutively from left to right. A seventh position shall be labeled calibrate.

3.5.2 **Amplifier.** The amplifier shall amplify the selected transducer signal for operation of the display and headset.

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3.5.2.1 **Frequency response.** The frequency response of the amplifier shall be flat to within plus or minus 3 dB from 30 to 15,000 Hz.

3.5.2.2 **Gain.** The amplifier shall have sufficient gain to produce a display reading (see 3.5.3.2) of 0 ± 1 dB (digital display reading of minus 30 ± 1 dB (see 3.5.3.1)) when the input from a transducer is that which corresponds to a pump vibration having an acceleration of minus 30 ± 1 dB relative to one g root-mean-squared (rms) in a direction monitored by that transducer (see 3.4) with step attenuator (see 3.5.2.5) set at minus 30 dB (meter only), when no filters (see 3.5.2.6) are in the circuit and the control-monitor is connected to the transducer as specified in 3.4.1. The specified requirement shall be for transducer signals which are sinusoidal in nature and for transducer signals which are random in nature.

3.5.2.3 **Signal-to-noise ratio.** The signal-to-noise ratio of the amplifier shall be not less than plus 10 dB for the signal specified in 3.5.2.2.

3.5.2.4 **Linearity.** The gain of the amplifier shall be linear within plus or minus 5 percent for signal inputs varying from that corresponding to a pump vibration of minus 40 dB relative to one g to that corresponding to a pump vibration of plus 30 dB relative to one g.

3.5.2.5 **Step attenuator.** A six-position attenuator switch shall be provided as a front panel control. This control shall provide 50 plus or minus 1 dB of attenuation in its extreme left-hand position, 40 plus or minus 1 dB in the next position, 30 plus or minus 1 dB in the next, 20 plus or minus 1 dB in the next, 10 plus or minus 1 dB in the next, and zero attenuation in its extreme right-hand position. The respective positions of this control shall be marked +20, +10, 0, −10, −20, and −30 dB. This control, in combination with the meter (see 3.5.3.2) provides the control-monitor with a dynamic range of 70 dB. A step attenuator is not allowed if a digital numeric display (see 3.5.3.1) is used except if required for the headset circuit (see 3.5.4), in which case it shall have no effect on the digital display.

3.5.2.6 **Filters.** The amplifier shall be provided with two high pass filters. A selector switch shall be provided on the front panel to select either one or neither of these filters. The insertion loss of these filters over their passbands shall not exceed 1 dB. One filter shall pass all frequencies above 1,000 Hz and shall suppress all frequencies below 500 Hz by at least 20 dB relative to the 1,000 Hz level. The other filter shall pass all frequencies above 5,000 Hz and shall suppress all frequencies below 2,500 Hz by at least 20 dB relative to the 5,000 Hz level.

3.5.3 **Display circuit.** If a digital numeric display circuit is used, the circuit shall conform to the requirements specified in 3.5,3.1. Otherwise, if an analog display circuit is used, the circuit shall conform to the requirements specified in 3.5.3.2.

3.5.3.1 **Digital numeric display circuit.** A digital numeric display circuit shall be provided to monitor the amplifier output. The numeric display shall provide a numeric range from minus 50 dB to plus 50 dB with an accuracy of plus or minus 1 dB. The display shall comply with the indicators and associated items requirements of MIL-E-16400.

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3.5.3.2 **Meter circuit.** A meter circuit shall be provided to monitor the amplifier output. The meter shall conform to the electrical, physical, and environmental requirements of MIL-M-10304. The meter shall be type MR36W---SPECR with a 90-degree scale extending form minus 10 dB at the left, through zero at the center, to plus 10 dB at the right. The scale shall be calibrated in 1 dB steps, equally spaced. The meter shall be mounted on the front panel of the control-monitor.

3.5.4 **Headset circuit.** A headset, type H-216/U, in accordance with MIL-STD-242, shall be furnished with each equipment. The control-monitor cabinet shall have an enclosed storage compartment for the headset and its cord. The control-monitor shall provide a cable equipped with a standard phono jack to connect the headset with the control-monitor. This cable shall be 1.83 m (6 (plus 1, minus 0) feet) in length including the jack. A gain control for the headset shall be provided on the front panel of the control-monitor. If a meter is used, an easily discernible signal shall be produced in the headset whenever the meter reading is zero and the gain control is at midposition, regardless of the setting of the step attenuator. If a digital numeric display is used, an easily discernible signal shall be heard for indication values between minus 50 dB and plus 50 dB by adjustment of the step attenuator and headset gain control. The display reading shall not be affected by any setting of the headset gain control nor by a short or open circuit in the head set cord.

3.5.5 **Calibration circuit.** A calibration circuit shall be provided with the calibration resistor in series with the transducer selected by the pump selector switch. The calibration circuit shall serve as a means of checking the condition of, and compensating for aging effects on, the amplifier. Indicator lights shall not be provided to show that the equipment is under test or to show that the equipment tests out satisfactorily. The calibration circuit shall be energized by placing the pump selector switch to the calibrate position.

3.5.5.1 **Frequency.** The calibration frequency shall be 400 plus or minus 40 Hz.

3.5.5.2 **Amplitude.** The amplitude of the calibration signal shall be adjustable by means of a locked potentiometer screwdriver adjustment, internally mounted. At approximately mid-position of this adjustment, the amplitude shall be such that the display reads 0 dB with the step attenuator set at its position marked zero (meter only) when the control-monitor and a transducer are connected for operation in accordance with 3.4.1. Test points shall be provided inside the cabinet for measuring the amplitude of the calibration signal.

3.6 **Mechanical.**

3.6.1 **Control-monitor.**

3.6.1.1 **Dimensional limitations.** The control-monitor cabinet shall be rectangular. Its depth, height, and width shall not exceed 25.40, 30.48, and 30.48 centimeters (cm) (10, 12, and 12 inches), respectively. The dimensions include the headset storage space (see 3.5.4).

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3.6.1.2 **Cable entrance plates.** Cable entrance plates shall be located in the top and bottom panels of the cabinet and at the rear of the cabinet. Each plate shall handle all of the external cabling required by the control-monitor.

3.6.1.3 **Mounting.** The control-monitor shall be constructed for bulkhead mounting without shock mounts.

3.6.1.4 **Operator operating position.** The control-monitor shall be constructed for standing, eye level operation.

3.6.1.5 **Weight.** The weight of the control-monitor shall not exceed 38 pounds (17.2 kilograms).

3.6.1.6 **Enclosure.** The enclosure shall be in accordance with MIL-STD-108. Degree of enclosure shall be dripproof (45 degrees) except that no opening shall be permitted.

3.6.2 **Transducers.**

3.6.2.1 **Dimensional limitations.** The vertical dimensions of each transducer, including the connector, shall not exceed 5.08 cm (2 inches). The horizontal dimensions shall not exceed a 5.08 by 12.70 cm (2 by 5-inch) rectangular area, excluding the connector receptacle (see 3.6.2.2). The connector shall cause the cable to lead away from the transducer on a 5.08 by 5.08 cm (2 by 2-inch) face and in a plane parallel to the transducer/pump interface. A straight connector receptacle in accordance with MIL-C-5015 shall be used.

3.6.2.2 **Transducer enclosure.** The transducer enclosure shall be watertight in accordance with MIL-E-2036 except that no special means shall be provided for draining condensed moisture. A watertight connector receptacle shall be integral with each transducer and shall not affect its watertightness.

3.6.2.3 **Mounting.** Transducers shall be constructed for mounting on a flat surface of a pump using four bolts (see 3.4.7). The bolt spacing shall be rectangular, 41.28 ± 0.13 and 92.08 ± 0.13 mm (1.625 ± 0.005 and 3.625 ± 0.005 inches) between centers.

3.7 **Electrical.**

3.7.1 **Supply line voltage and frequency.** The equipment shall operate from a voltage supply of 115 volts (V) alternating current (ac) plus or minus 10 percent, 60 Hz plus or minus 5 percent, single-phase, ungrounded, type I shipboard power as specified in MIL-STD-1399, section 300, and shall conform to the user equipment requirements specified in this standard.

3.7.2 **Insulation resistance.** Insulation resistance at 25 degrees Celsius (°C) between electrically isolated circuits, and between circuits and ground, shall be at least 10 megohms.

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3.7.3 **Electrical power ON-OFF switch.** A switch for disconnecting from all electrical power systems shall be mounted on the control-monitor front panel and the function clearly labeled. The switch shall break all power conductors.

3.7.4 **Power indicator lamp.** A green power indicator lamp shall be mounted near the switch to indicate when the control-monitor is energized. The lamp shall be connected to the load side of the switch and across the input power conductors. The indicator shall conform to MIL-STD-454, requirement 50.

3.7.5 **Automatic bus transfer.** The control-monitor shall continue to operate satisfactorily with no equipment damage following a 50 millisecond interruption of power during transfer from one power supply to another.

3.7.6 **Circuit breakers and fuses.**

3.7.6.1 **Circuit breakers, fuses, and fuseholders.** Circuit breakers, fuses, and fuseholders shall conform to their respective requirements specified in MIL-E-16400. Both sides of the power input shall be protected.

3.7.6.2 **Overcurrent device location.** A fuse or an overcurrent trip unit of a circuit breaker shall be connected in series with each ungrounded conductor as determined by the National Electrical Code (see NFPA 70) for grounded source power and shall be connected on the load side of the ON-OFF power switch. The ON-OFF switch may be the circuit breaker.

3.7.7 **Power connections.** Control-monitor power connectors for 115 Vac input power and interconnecting alternating current power cable connectors between control-monitors shall have only three pins designated A, B, and C. The connector shall be in accordance with MIL-C-5015. Pin assignment and conductor color code within the control-monitor shall be as follows:

Pin designation	Conductor assignment	Color code
A	115 Vac return	White or light blue
B	Safety ground	Green or green with yellow stripe
C	115 Vac high (hot)	Black or brown

Black or brown and white or light blue color code shall be maintained from the input connector pins to all components having the same voltage and frequency as the input power.

3.7.8 **Leakage current (equipment).** The leakage current shall conform to the leakage current (equipment) requirement of MIL-E-16400. The leakage current shall not exceed 5 milliamperes (mA).

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3.7.9 Dielectric withstanding voltage. The dielectric withstanding voltage shall conform to the dielectric withstanding voltage requirement of MIL-E-16400.

3.7.10 Electromagnetic environmental effects (E^3). The equipment shall conform to the requirements of MIL-STD-461 for class A4 equipment. Requirements CE01, CE03, CS01, CS02, CS06, RE01, RE02, RS01, and RS02 apply.

3.7.11 Warm-up time. The time for the equipment to stabilize within the specified requirements shall be less than 30 minutes.

3.8 Thermal design. The thermal design shall be in accordance with the thermal design requirement of MIL-E-16400.

3.9 Reliability. The upper test mean-time-between failures (MTBF) (O_0 as defined by MIL-STD-781) shall be 4500 hours. The lower test MTBF (O_1 as defined by MIL-STD-781) shall be 1500 hours. Failure is the inability to conform to any of the performance requirements of this specification.

3.10 Maintainability. Maintainability shall be as specified in 3.10.1 through 3.10.3 (see 6.3).

3.10.1 Quantitative corrective maintenance. The equipment shall have a mean-corrective-maintenance-time (M_{ct}) not exceeding 1.0 hour and a maximum-corrective-maintenance-time (M_{maxct}) (95th percentile) not exceeding 2.0 hours when corrective maintenance is accomplished at the organizational level. The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. The equipment is considered to be in a corrective maintenance mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

3.10.2 Test points criteria. Test points shall be as follows:

a. *Signal quality and circuit performance.* Test points, test jacks, or both, shall be provided to permit the injection of signals and the monitoring of signals at the input/output (I/O) terminals of the subassembly.

b. *Fault isolation.* Test points, test jacks, or both, shall be provided to permit the injection or measurement of signals at I/O terminals of the subassembly for the purpose of isolating all faults to electrical functions (that is, integrated circuit (IC), filter, mixer, voltage controlled oscillator (VCO); and so forth). Test points and test jacks shall provide a means for performing fault isolation on the subassembly with the subassembly removed form the prime equipment.

c. *Alignment.* Test points, test jacks, or both, shall be provided to permit alignment of the equipment and equipment assemblies and subassemblies.

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3.10.3 **General purpose electronic test equipment (GPETE).** GPETE to be used at all levels of maintenance shall be selected from the standard and substitute standard equipments listed in MIL-STD-1364.

3.11 **Parts, materials, and processes.** Parts, materials, and processes shall be as specified in 3.11.1 through 3.11.4.

3.11.1 **Applicable MIL-E-16400 requirements.** All parts, materials, and processes shall be in accordance with MIL-E-16400 with the following exceptions:

a. Tube shields
b. Transmission lines and related radio frequency (RF) parts.

3.11.2 **Capacitors.** Capacitors shall be derated at least 50 percent; for example, the actual maximum peak voltage that the component experiences shall be not greater than 0.5 times the specification voltage.

3.11.3 **Resistors.** Resistors shall be derated at least 50 percent; for example, the actual wattage of the component in the circuit shall be not greater that 0.5 times the specification wattage rating as calculated for the ambient condition.

3.11.4 **Stress derating.** The contractor shall establish and maintain specific criteria and methods acceptable to the Government for stress derating of electrical and mechanical parts. Criteria shall be stated in terms of applied versus rated voltage, power, junction temperature, strain, and so forth, as applicable for each part type. Practices shall be implemented which will ensure that the established derating criteria are adhered to during the design process.

3.12 **Identification and marking.** Identification and marking shall conform to the identification and marking requirements of MIL-E-16400.

3.13 **Power supply protection.** Fault conditions ranging from open circuits to short circuits shall cause no damage to the power supply.

3.14 **Printed circuit assembly protection.** With power applied to the control-monitor, removal or insertion of printed circuit assemblies shall not damage or degrade any part of the control-monitor. Each assembly shall be keyed to prevent insertion into the wrong connector and to prevent reversal of insertion.

3.15 **Human error design protection.** The construction of the control-monitor shall prevent damage as a result of operator errors involving procedures in control sequencing, adjustments, and in responding to fault indicators.

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3.16 Mechanical hazards. All accessible corners shall be rounded. Exterior doors, hinged covers, and drawers shall have automatic latch and quick release devices which shall be actuated before they can be opened and stops to hold them in the open position for each maintenance configuration. Control-monitor construction shall include provisions to prevent accidental derailing, opening, and pulling out of drawers of rack-mounted equipment; and shall include means to prevent accidental dropping of exterior covers and panels during removal and attachment.

3.17 Human engineering. Human engineering shall comply with the human engineering operational and maintenance design criteria of MIL-STD-1472.

3.18 Spare connections. Printed wiring board (PWB) connectors shall have at least 10 percent spare connections. Each spare connection shall be connected by printed wiring to a separate spare pad on the PWB. The wiring pattern for the spare terminals shall not be beneath the connector. Space and component pads (not the same pads connected to the spare connectors) for addition of spare components shall be provided on each circuit board. The number of additional components allowed for shall be at least 10 percent of each different component. Different component types (for example, resistors, capacitors, diodes) constitute different components for the purpose of spare component pads. Different components which have the same mounting configuration (1/2 wart resistors, 2 watt resistors, and so forth) shall have separate spare component pads. At least 10 percent means the next integer higher than or equal to 10 percent of the total of that component type. In addition, the number of pads shall be maximized. The passive component spare pads should be arranged to allow mounting of either resistors or capacitors. The distribution of spare pads shall be spaced to permit addition of parts by future modifications. In the event that a circuit change is required on completed circuit boards, the change shall be accomplished by cutting printed conductors and adding wires which would be soldered in the extra plated through holes provided for that purpose. Spare component mounting pads shall have two additional holes (other than the mounting holes for adding the new part) for interconnecting wiring or components. Printed wiring to semiconductor devices other than diodes shall contain at least two plated holes per circuit connection. Where ICs or other semiconductor devices other than diodes are mounted on the PWB, provision shall be made for mounting at least one more of each type of device used on the board. In this instance, "type" refers to mounting configuration. It is not permissible to meet the requirement for extra IC mounting provisions by superimposed circuit board hole patterns. Connections shall be made to plated-through holes and all spare holes in the PWB shall be through plated. ICs and other semiconductor devices other than diodes shall be positioned to allow the breaking of connections between them and to allow the installation of components or wires between them. The PWB shall contain uncoated pad test points at the input and output of each major active component. The uncoated area shall be limited just to the test pad.

3.19 Transportability. The equipment shall withstand the environmental problems resulting from normal transportation and handling without damage or calibration change. Calibration and alignment changes that can be corrected following normal installation methods by using only the front operator's controls will be acceptable.

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3.20 **Workmanship.**

3.20.1 **General workmanship.** Workmanship shall conform to MIL-STD-454, requirement 9.

3.20.2 **Workmanship screen.** Equipment shall withstand a defect detection vibration screen (see 4.5.3.1) of random type vibration at 0.04 g^2/Hz plus or minus 3 dB from 80 to 350 Hz and temperature cycling (see 4.5.3.2) with operation at the maximum and minimum operating temperatures and dwell time at the nonoperating temperature extremes.

3.20.3 **Soldering and soldering processes.** Soldering and soldering processes shall conform to MIL-STD-454, requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Quality assurance terms and definitions.** Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. Unless otherwise specified, three units shall be required for first article inspection. First article inspection shall include tests specified in table II.

TABLE II. *First article inspection.*

Examination or test	Requirement	Test
Dimension limitations	3.6.1.1 and 3.6.2.1	4.5.1
General workmanship	3.20.1	4.5.1
Maintainability	3.10	(see 6.3)
Parts, materials, and processes	3.11	4.5.1
Identification and marking	3.12	4.5.1
Reliability	3.9	4.5.6
Operating test	3.4 through 3.4.4.3; 3.5 through 3.5.5.2; and 3.13 through 3.15	4.5.2
Humidity	3.3.2	4.5.10
Shock	3.3.8	4.5.5
Inclination	3.3.5	4.5.11
Circuit breakers and fuses	3.7.5	4.5.1
Thermal design	3.8	4.5.1
Enclosure	3.6.1.6	4.5.4
Power	3.7.7	4.5.1
Dielectric withstanding voltage	3.7.9	4.5.12
Transducer enclosure	3.6.2.2	4.5.16
Insulation resistance	3.7.2	4.5.1
Supply line voltage and frequency	3.7.1	4.5.13
Warm-up time	3.7.11	4.5.9
E[3]	3.7.10	4.5.7
Leakage current	3.7.8	4.5.8
Workmanship screen	3.20.2	4.5.3
Temperature conditions	3.3.1	4.5.17
Pressure conditions	3.3.4	4.5.14
Incident nuclear radiation	3.3.3	4.5.15
Materials	3.4.5	4.5.1
Cable connector	3.4.6	4.5.1
Transducer mounting hardware	3.4.7	4.5.1
Mounting	3.6.1.3 and 3.6.2.3	4.5.1
Electrical power ON-OFF switch	3.7.3	4.5.1
Power indicator lamp	3.7.4	4.5.1
Mechanical hazards	3.16	4.5.1
Human engineering	3.17	4.5.18

4.4 Quality conformance inspection. Quality conformance inspection shall be made on every equipment offered for delivery. Quality conformance inspection shall include the tests and examinations specified in table III.

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TABLE III. *Quality conformance inspection.*

Inspection	Requirement	Test
Dimensional limitations	3.6.1.1 and 3.6.2.1	4.5.1
General workmanship	3.20.1	4.5.1
Workmanship screen	3.20.2	4.5.3
Parts, materials, and processes	3.11	4.5.1
Identification and marking	3.12	4.5.1
Reliability[1]	3.9	4.5.6
Operating test	3.4 through 3.4.4.3; 3.5 through 3.5.5.2; and 3.13 through 3.15	4.5.2
Dielectric withstanding voltage	3.7.9	4.5.12
Insulation resistance	3.7.2	4.5.1
Leakage current	3.7.8	4.5.8

[1]When first article inspection is waived.

4.5 Inspection method.

4.5.1 Examinations and tests. Examinations and tests specified in table II and table III shall be conducted in accordance with the applicable test of MIL-E-16400, except as specified in 4.5.2 through 4.5.15 (see 6.3).

4.5.2 Operating test. The equipment shall be energized and subjected to an operating test to verify conformance to the requirements of 3.4 through 3.4.4.3; 3.5 through 3.5.5.2; and 3.13 through 3.15 (see 6.3).

4.5.3 Workmanship screen. Vibration and temperature cycling shall be performed on each control-monitor as specified in 4.5.3.1 and 4.5.3.2. Vibration shall be performed prior to temperature cycling.

4.5.3.1 Vibration. Vibration shall be as follows:

a. The vibration test may be performed at the module, drawer or end item level. All the hardware including cables and connectors shall be exposed to vibration.

b. The vibration shall be random, or subject to contracting activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. Test items shall be hardmounted (without shock isolators) and subjected to the vibration conditions of figure 1.

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c. Input vibration levels shall be measured at the mounting points of the item under test. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth not greater than 10 Hz for the vibration frequencies up to 500 Hz, and 100 Hz for vibration frequencies above 500 Hz, shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the rms acceleration level.

d. The control-monitor need not be energized during vibration. Failures occurring during screening shall be corrected and the vibration resumed.

e. The headset shall be in the stowed position during the vibration test. Post vibration inspection and tests shall consist of the quality conformance inspection of 4.4.

4.5.3.2 **Temperature cycling.** Each control-monitor shall be subjected to 10 cycles of the temperature curve shown on figure 2. The temperature rate of change shall be not less than 5 °C per minute. Equipment power shall be turned on and off at the indicated times. The control-monitor shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part reaches 80 percent of the chamber temperature. When failures occur, the equipment shall be reworked and the cycling continued for a cumulative total of 10 cycles.

4.5.4 **Enclosure.** The control-monitor shall be tested to verify conformance to the requirements of 3.6.1.6.

4.5.5 **Shock test.** The control-monitor shall be tested to verify conformance to the requirements of 3.3.8. During the shock test, the headset shall be in the stowed position.

4.5.5.1 **Failure under shock test.** The equipment shall be considered as having failed the shock test if it fails to perform any of its intended functions or indicates a continuous incorrect display reading greater than plus or minus 3 dB. Transient vibrations in the output signals of the equipment during high impact shock shall be of a minimum amplitude and duration.

4.5.5.2 **Post-shock inspection.** Post-shock inspection shall consist of the quality conformance inspection of 4.4. Shock tested equipment shall not be installed on board ship.

4.5.6 **Reliability testing.** A sample of at least three first article units, or three units from the first production lot should first article be waived, shall be subjected to reliability testing in accordance with test plan VC of MIL-STD-781 to demonstrate compliance with the requirements of 3.9. Environmental conditions shall be as specified herein.

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4.5.6.1 **Temperature.** Equipment shall be tested at an environmental temperature of 50 °C, 5 °C, and 0 °C.

4.5.6.2 **Vibration.** A 4- to 50-Hz to 4-Hz vibration sweep shall be performed throughout the reliability test with vibration displacement (table amplitude) as defined by MIL-STD-167-1. The duration of each complete sweep shall be approximately 15 minutes.

4.5.6.3 **Moisture.** Once per test day, moisture shall be injected into the test chamber. The moisture level need not be strictly controlled, but shall be sufficient to cause visible condensation in the chamber and on the equipment being tested.

4.5.6.4 **Operating duty cycle.** Equipments shall be operating and functioning under normal operating mode conditions throughout reliability testing. Normal operating mode shall be those inputs (power, audio signals, and so forth) which would simulate actual operational use. Only actual operating time shall be used for establishing an accept/reject decision.

4.5.6.5 **Failure.** The contracting activity shall be notified within 2 working days of any failure during reliability testing. Classification of failures shall be determined by the contracting activity in accordance with the failure actions requirement of MIL-STD-781.

4.5.6.6 **Rejected equipment.** In the event equipment is rejected as a result of reliability testing, the contractor shall immediately notify the contracting activity of the causes for such rejection and propose a plan for correction of the deficiencies. When such corrections have been approved by the contracting activity and retrofitted in the test units, the reliability test shall be reperformed as specified herein.

4.5.6.7 **Reinspection of conforming reliability sample units.** Unless otherwise specified in the contract (see 6.2), sample units which have been subjected to and passed reliability tests may be accepted on the contract, provided they are resubjected to and pass production inspection.

4.5.7 **EMC test.** Conformance to the requirements of 3.7.10 shall be in accordance with MIL-STD-462.

4.5.8 **Leakage current test.** The leakage current test shall be as specified in 4.5.8.1 and 4.5.8.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE CONTROL-MONITOR DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

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4.5.8.1 **Control-monitor connections.** Each control-monitor directly connected to an external power source and units deriving power from the control-monitor shall be placed on an insulated surface. All safety ground conductors between the control-monitor and units deriving power from the control-monitor shall be intact. The safety ground conductor between the control-monitor under test and the source power shall be opened during the test. OBSERVE WARNING STATEMENT. The control-monitor shall be connected as shown on figures 3 if it is connected to single-phase source power. When direct current (dc) measurements are performed, do not reverse polarity unless the equipment has reverse polarity protection.

4.5.8.2 **Measurement.** Leakage current shall be measured on the control-monitor in its normal operating configuration. Control-monitor controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the control-monitor is designed. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available on figure 3. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.9 **Warm-up time.** The control-monitor shall be energized for exactly 30 minutes after which the unit shall be calibrated. This calibration shall be checked for 8 hours and shall remain within plus or minus 10 percent of the original calibration.

4.5.10 **Humidity.** Humidity tests shall be in accordance with method 507.2, procedure III of MIL-STD-810, except that:

a. The temperature-humidity cycle shall be in accordance with figure 4.

b. In step 2, the internal chamber temperature shall be raised to 50 °C (122 degrees Fahrenheit (°F)).

c. In step 6, the steps shall be repeated for a total of 5 cycles (not less than 120 hours).

d. In step 7, the test shall be performed after the third and fifth cycle.

4.5.11 **Inclination.** The equipment shall be inclined for at least 5 minutes in each of the following positions:

a. 60 degrees forward
b. 60 degrees backward
c. 60 degrees to the left
d. 60 degrees to the right.

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4.5.12 **Dielectric withstanding voltage test.** Dielectric withstanding voltage test shall be as specified in MIL-E-16400 except that the rms test voltage shall be twice the rotated circuit voltage plus 1000 V.

4.5.13 **Supply voltage and frequency.** The equipment shall be operated at the ambient temperatures and the supply voltages and frequencies specified in table IV until a stable operating condition has been reached. The performance characteristics, such as the accuracy of the output and of each set/trip point and any other equipment function which could be affected, shall be determined for each condition.

TABLE IV. *Supply voltage and frequency test.*

Ambient temperature	Voltage (percent)	Frequency (percent)
5 ± 5 °C	100	100
	110	95
	110	105
	90	105
	90	95
25 ± 5 °C	100	100
	110	95
	110	105
	90	105
	90	95
50 ± 5 °C	100	100
	110	95
	110	105
	90	105
	90	95

4.5.14 **Pressure.** The control-monitor and transducers shall operate under the conditions of 3.3.4 without degradation.

4.5.15 **Incident radiation.** The control-monitor and transducers shall operate under the conditions of 3.3.3 without degradation.

4.5.16 **Transducer enclosure.** The transducer enclosure shall be tested to verify conformance to the requirements of 3.6.2.2.

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4.5.17 **Temperature.**

4.5.17.1 **Low temperature.** The low temperature test shall be in accordance with method 502.2, procedure I and II of MIL-STD-810. The storage temperature shall be maintained for a period of not less than 12 hours after stabilization.

4.5.17.2 **High temperature.** The high temperature test shall be in accordance with method 501.2, procedures I and II of MIL-STD-810. For procedure I the storage cycle shall consist of one cycle for a total of 24 hours. For procedure II the temperature exposure shall be constant.

4.5.18 **Human engineering.** The equipment shall be tested to verify conformance to the requirements of 3.17.

4.6 **Inspection of packaging.** Sample packs, and the inspection of the packaging and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 **General.**

5.1.1 **Navy fire-retardant requirements.**

a. *Treated lumber and plywood.* When specified (see 6.2), all lumber and plywood including laminated veneer materials used in shipping container and pallet construction members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B	–	Type II – weather resistant Category 1 – general use
Level C	–	Type I – non-weather resistant Category 1 – general use

b. *Fiberboard.* Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements of PPP-F-320 and amendments thereto.

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5.2 **Packaging requirements.** The packaging (preservation, packing, and marking) requirements shall be in accordance with MIL-E-17555 for the level of preservation (A, B, C, or commercial), level of packing (A, B, C, or commercial) and marking including bar coding and other packaging acquisitioning options therein as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The equipment described in this specification is intended for shipboard use, as follows:

a. Type I equipment is used to monitor the operation of six pumps. Each pump is monitored by one transducer. The signal from each transducer is representative of pump vibration in three perpendicular directions in space.

b. Type II equipment is used to monitor the vibration of six pumps. Each pump is monitored by one transducer. Two signals are provided by each transducer, corresponding to vibration in two directions. In normal operation these signals are connected in parallel by jumpers in the transducer cabling. For special testing purposes the output of each accelerometer can be individually monitored by removing the jumpers.

c. Type III equipment is used to monitor the vibration of four pumps. Although three accelerometer outputs are available at each transducer only two are normally connected. Two signals from each transducer are paralleled by jumpers in the transducer cabling to provide one signal per transducer to the control-monitor. For special testing the output of each accelerometer can be individually monitored by removing jumpers.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. Type required (see 1.2)

d. When first article is required (see 3.1)

e. When reliability test sample units may not be accepted on the contract (see 4.5.6.7)

f. When fire-retardant treated materials are not required (see 5.1.1a and b)

g. Level of packaging and packing, and other packaging options required (see 5.2).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.10	DI-R-7103	Maintainability Program Plan	–
4.1.1	DI-E-2121	Certification of Compliance	–
4.5.1 and 4.5.2	DI-T-2072	Test Reports	–

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.6 Subject term (key word) listing.

Accelerometer
Control-monitor
Pump, cooling water
Transducer
Vibration

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 6665-N554)

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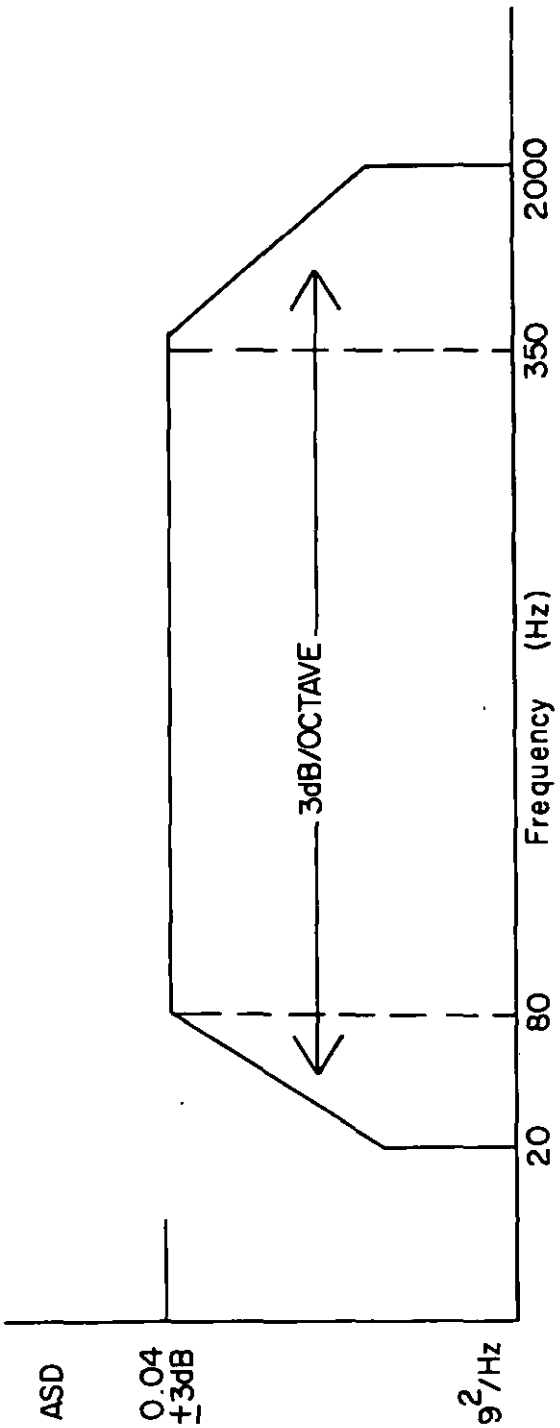


FIGURE 1. *Random vibration curve.*

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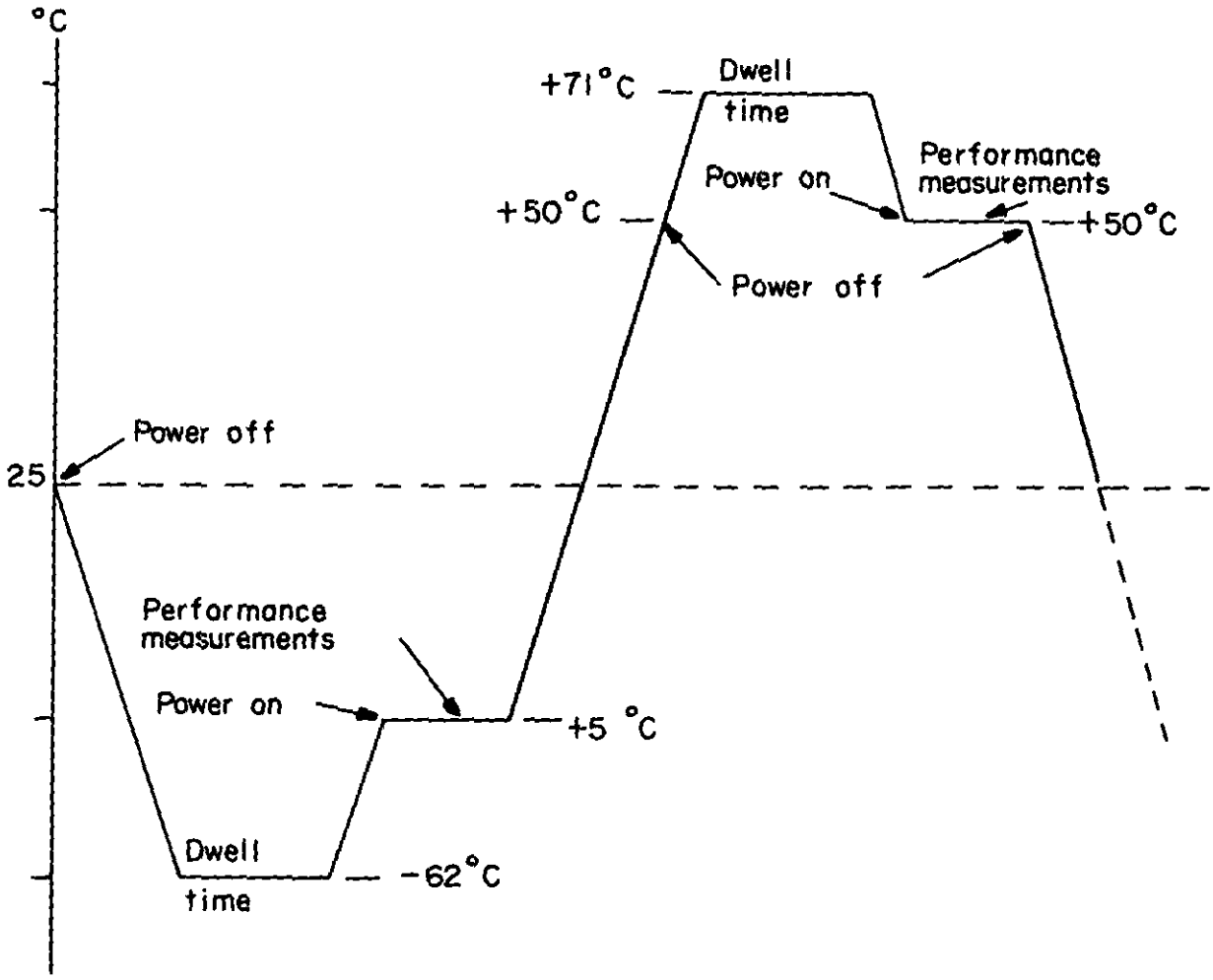


FIGURE 2. *One cycle of temperature curve (not to scale).*

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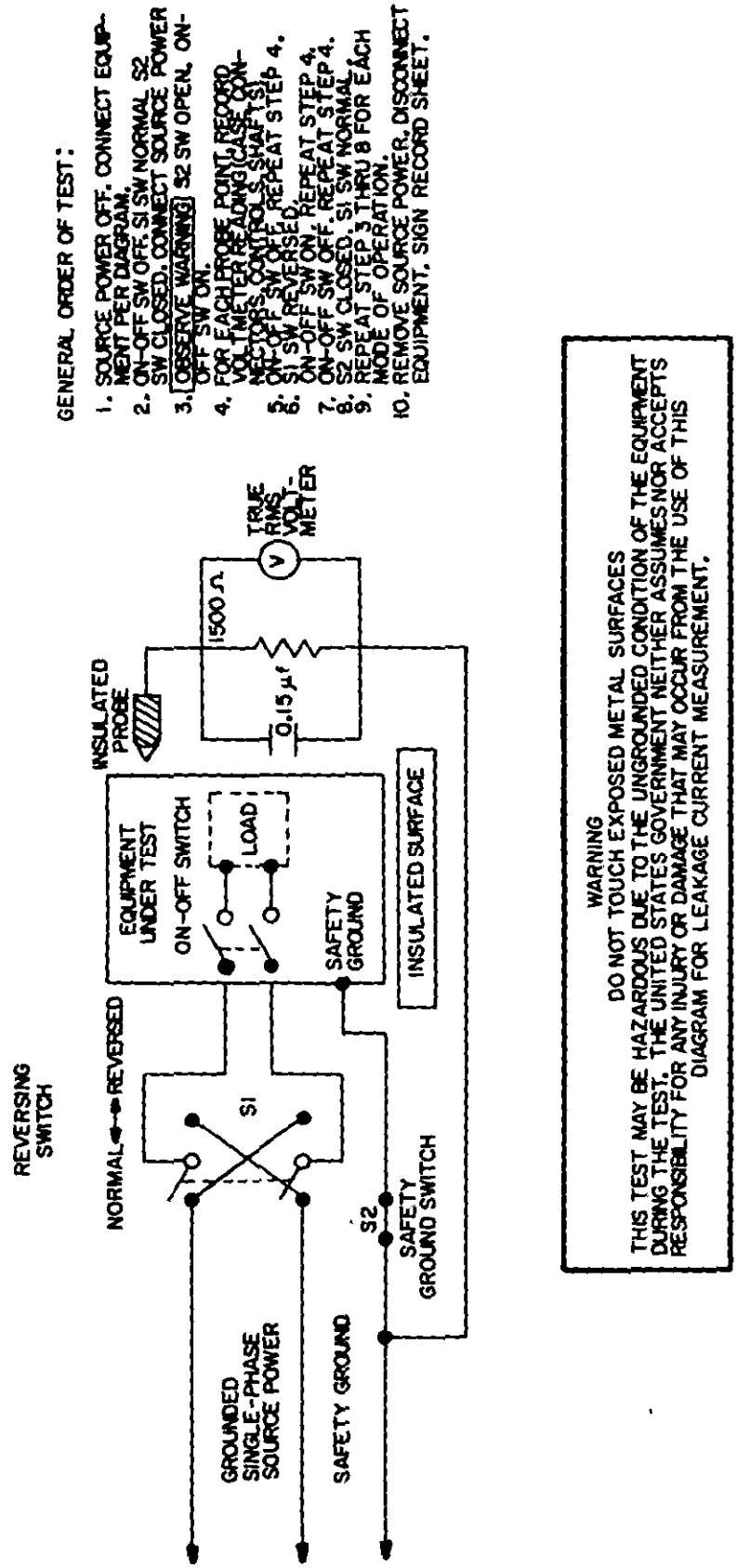
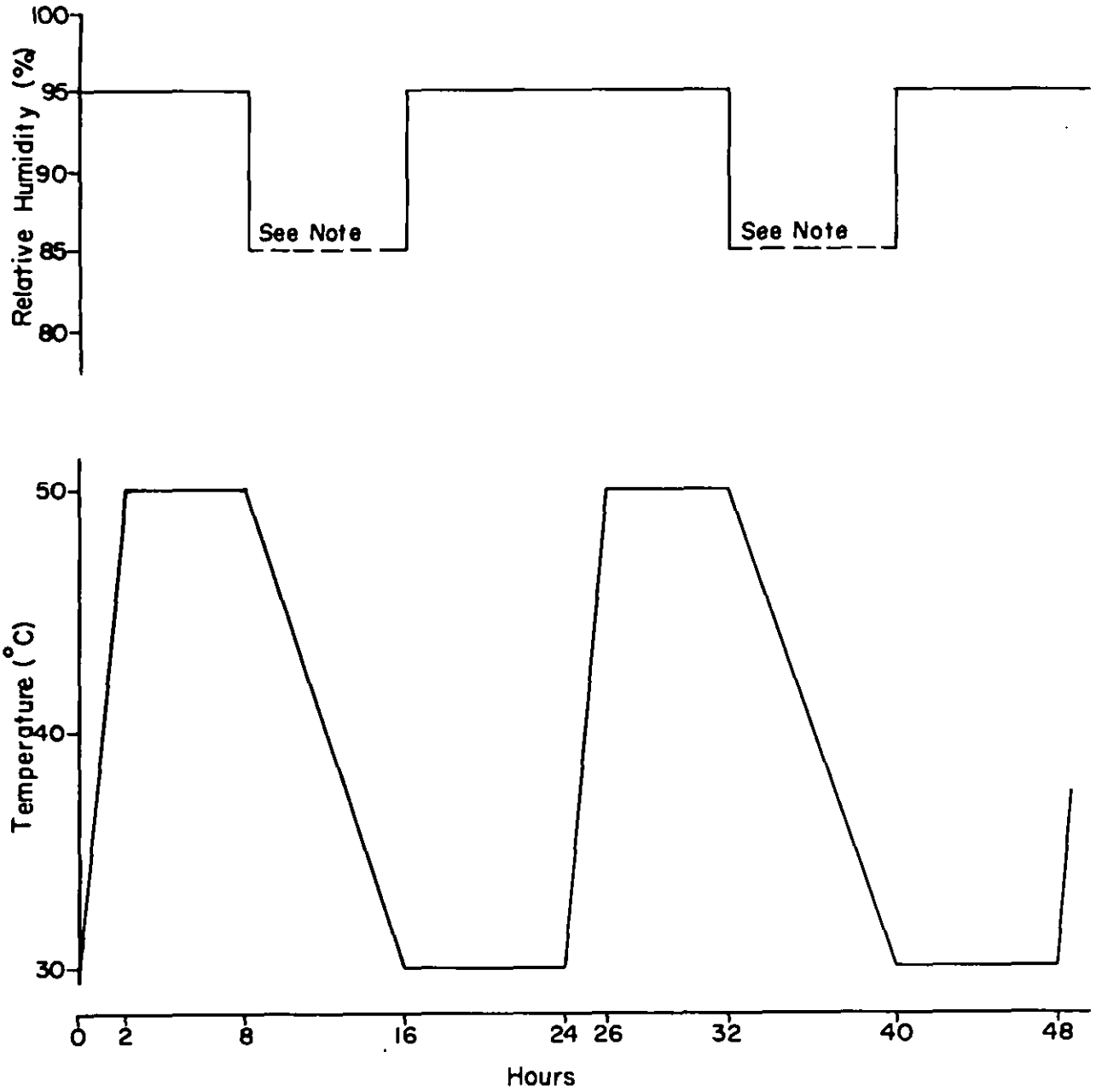


FIGURE 3. *Single-phase test diagram for leakage current measurement.*

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Note: Relative humidity maintained above 85 percent during temperature drops.

FIGURE 4. *Aggravated temperature-humidity cycles.*

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-28762C(SH)	1 August 1990

3. DOCUMENT TITLE: Monitor, Pump Noise AN/WSQ-3 (METRIC)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME Mr. Kelvin W. Predmore	b. TELEPHONE (Include Area Code) (1) Commercial (703) 692-7385	(2) AUTOVON 322-7385
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 — Previous editions are obsolete. — 198/290

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